

1.1.1

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Ecuación parabólica:

$$y(t) = y_0 + v_0 \sin \theta t - \frac{1}{2} g t^2 \quad (1)$$

$$y_0 = 0$$

$$x = v_x t$$

$$x = v_0 \cos \theta t \rightarrow t = \frac{x}{v_0 \cos \theta} \quad (2)$$

(2) en (1):

$$h = x \tan \theta - \frac{g x^2}{2 v_0^2 \cos^2 \theta}$$

Despejar v_0 :

$$x \tan \theta - h = \frac{g x^2}{2 v_0^2 \cos^2 \theta}$$

$$\Leftrightarrow 2 v_0^2 \cos^2 \theta = \frac{g x^2}{x \tan \theta - h}$$

$$v_0 = \sqrt{\frac{g x^2}{2(x \tan \theta - h) \cos^2 \theta}} \quad (3)$$

Ahora toca $N'_0(\theta) = 0$.

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$$N_0 = x \sqrt{\frac{g}{2}} \left[\underbrace{(x \tan \theta - h) \cos^2 \theta}_{f(x)} \right]^{-1/2}$$

$$N'_0 = \frac{-x}{2} \sqrt{\frac{g}{2}} \cdot f(x) \cdot f'(x) = 0$$

De (3) $f(x) \neq 0$.

Entonces

$$f'(x) = 0$$

$$\Leftrightarrow (x \sec^2 \theta)(\cos^2 \theta) + (x \tan \theta - h)(2 \cos \theta)(-\sin \theta) = 0$$

$$\Leftrightarrow x - 2(x \tan \theta - h) \cos \theta \sin \theta = 0$$

$$\Leftrightarrow x = 2x \tan \theta \cos \theta \sin \theta - 2h \cos \theta \sin \theta$$

$$\Leftrightarrow x = 2x \sin^2 \theta - h \sin(2\theta)$$

$$\Leftrightarrow x = x(1 - \cos(2\theta)) - h \sin(2\theta)$$

$$\Leftrightarrow x = x - x \cos(2\theta) - h \sin(2\theta)$$

$$\Leftrightarrow x \cos(2\theta) = -h \sin(2\theta)$$

$$\Leftrightarrow \frac{x}{h} = -\tan(2\theta) \quad (4)$$

$$\Leftrightarrow \frac{x}{h} = \tan(\pi - 2\theta) \rightarrow \boxed{\theta = \frac{\pi}{2} - \frac{1}{2} \arctan\left(\frac{x}{h}\right)}$$

[1.2.]

Tenemos

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$$n_0 = \sqrt{\frac{gx^2}{(2l \tan \theta - h) \cos^2 \theta}} \quad (3)$$

$$\theta_{\min} = \frac{\pi}{2} - \frac{1}{2} \arctan(x/h) \quad (4)$$

No vamos a meter (4) en (3) porque es una pelada. Queda un bicho bien horrible. Así que vamos a sacar los prohibidos.

$$\text{Sea } \arctan(x/h) = \phi \quad (5)$$

Entonces (5) en (4):

$$\theta_{\min} = \frac{\pi - \phi}{2}$$

Queremos expresiones para $\cos(\theta_{\min})$ y $\tan(\theta_{\min})$ que no tengan la $\arctan$

$$\begin{aligned} \cos(\theta_{\min}) &= \cos\left(\frac{\pi}{2} - \frac{\phi}{2}\right) \\ &= \cancel{\cos(\pi/2)} \cos(\phi/2) + \cancel{\sin(\pi/2)} \sin(\phi/2) \\ &= \sin(\phi/2) \\ &= \sqrt{\frac{1 - \cos \phi}{2}} \quad (6) \end{aligned}$$

$$\sin(\theta_{\min}) = \sin\left(\frac{\pi}{2} - \frac{\phi}{2}\right)$$

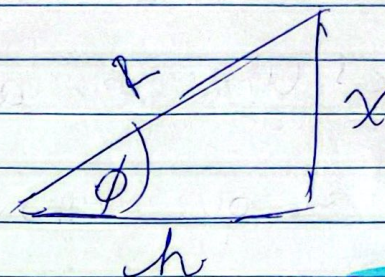
$$= \sin\left(\frac{\pi}{2}\right)\cos\left(\frac{\phi}{2}\right) - \sin\left(\frac{\phi}{2}\right)\cos\left(\frac{\pi}{2}\right)$$

$$= \cos\left(\frac{\phi}{2}\right)$$

$$= \sqrt{\frac{1 + \cos \phi}{2}} \quad (7)$$

De (5):

$$\tan \phi = \frac{x}{h}$$



$$R = (h^2 + x^2)^{1/2}$$

$$\cos \phi = \frac{h}{R} \quad (8)$$

$$\sin \phi = \frac{x}{R}$$

(3) en (6) y (7)

$$\cos(\theta_{\min}) = \sqrt{\frac{1 - \frac{h}{R}}{2}} \quad (9)$$

$$\sin(\theta_{\min}) = \sqrt{\frac{1 + \frac{h}{R}}{2}} \quad (10)$$

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(9) y (10) en el denominador (sin la raíz)
de (3):

$$\Leftrightarrow \left(1 - \frac{h}{R}\right) \left(x \left[\frac{1 + h/R}{1 - h/R} \right]^{1/2} - h \right)$$

$$\Leftrightarrow \left(\frac{R-h}{R} \right) \left(\frac{x(R+h)^{1/2} - h(R-h)^{1/2}}{(R-h)^{1/2}} \right)$$

$$\Leftrightarrow \frac{1}{R} (R-h)^{1/2} \left(x(R+h)^{1/2} - h(R-h)^{1/2} \right)$$

$$\Leftrightarrow \frac{1}{R} \left\{ x [(R-h)(R+h)]^{1/2} - h(R-h) \right\}$$

$$\Leftrightarrow \frac{1}{R} \left\{ x \underbrace{(R^2 - h^2)}_{x^2}^{1/2} - Rh + h^2 \right\}$$

$$\Leftrightarrow \frac{1}{R} \left\{ x^2 - Rh + h^2 \right\}$$

$$\Leftrightarrow \frac{1}{R} \left\{ R^2 - Rh \right\}$$

$$\Leftrightarrow R - h \quad (11)$$

(11) en (3)

$$v_0 = \sqrt{\frac{gx^2}{R-h}}$$

$$\Rightarrow v_0 = \sqrt{\frac{gx^2 (R+h)}{(R-h) (R+h)}}$$

$$= \frac{gx^2 (R+h)}{R^2 - h^2}$$

$$= \sqrt{\frac{gx^2 (R+h)}{x^2}}$$

$$= \sqrt{g(R+h)}$$

$$v_0 = \sqrt{g(\sqrt{x^2 + h^2} + h)}$$